
Learning an exact diffusion denoiser with thirteen parameters

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Abstract

1 The exact score of a diffusion model is unavailable for essentially every data dis-
2 tribution of interest, which makes it impossible to separate a network’s approx-
3 imation error from the intrinsic difficulty of the score it is approximating. We
4 study a data model where the exact score *is* computable: a first-order Markov
5 chain with continuous, non-Gaussian innovations. Coordinatewise noising leaves
6 the posterior a chain, chains are trees, and sum-product belief propagation on a
7 tree is exact — so BP returns the Bayes-optimal denoiser at every noise level. We
8 then remove the assumption that the chain is known and estimate its transition
9 kernel by expectation maximisation through exact BP. Three facts make this effi-
10 cient: the E-step is exact rather than variational, it compresses into a single matrix
11 of expected transition statistics, and Fisher’s identity supplies the exact gradient
12 of the marginal likelihood from one BP pass, so nothing is differentiated through
13 the recursion. On a Laplace chain, a 13-parameter estimator fitted to 32 noisy
14 sequences attains a lower score error than a 25,248-parameter denoising network
15 fitted to 4096 — a factor of more than 128 in data. We report the comparison
16 against a locality-respecting architecture as well, where the margin is $4.1\times$ and
17 *grows* with data rather than closing, and we state the cost: BP inference is two
18 orders of magnitude slower per evaluation.

19 1 Introduction

20 Diffusion models are trained by denoising regression. With $x = \alpha_t a + \sqrt{\Delta_t} z$ and $z \sim \mathcal{N}(0, I)$,
21 the objective $\mathbb{E}\|f_\theta(x, t) - a\|^2$ is minimised by the posterior mean $\mathbb{E}[a \mid x, t]$, from which the score
22 follows by Tweedie’s identity,

$$s(x, t) = -\frac{x - \alpha_t \mathbb{E}[a \mid x, t]}{\Delta_t}, \quad \alpha_t = e^{-t}, \quad \Delta_t = 1 - e^{-2t}. \quad (1)$$

23 There is a well-known tension in what this objective can be attaining. Its global minimiser over
24 all measurable functions is the *empirical* score — the score of a Gaussian mixture centred on the
25 training points — and a model that attained it would reproduce its training set and nothing else. That
26 trained diffusion models instead generalise is a central open question, and progress on it requires data
27 distributions for which the *exact* score is known, so that “the network learned the wrong score” can
28 be distinguished from “this score is hard”. The available such families are essentially Gaussians and
29 mixtures of Gaussians, which are too structureless to be informative about how a network exploits
30 structure.

31 We use a first-order Markov chain with continuous, non-Gaussian innovations. It is a minimal
32 model of the temporal coherence that motivates diffusion for video and other sequential data, and it
33 has a property that Gaussian families do not: the score is genuinely nonlinear in x , while remaining

34 exactly computable. The reason is structural rather than analytic. Coordinatewise noising multiplies
 35 the prior by a product of sitewise likelihood factors, which reweights node potentials without adding
 36 edges, so the posterior is again a chain; a chain is a tree; and sum-product belief propagation is exact
 37 on a tree. BP therefore returns the Bayes-optimal denoiser — not an approximation to it — at every
 38 noise level.

39 Our contributions are:

- 40 • **An exact denoiser for a non-Gaussian, correlated data model** (§2), together with the
 41 observation that the Gaussian message closure is not merely approximate but is *exactly* the
 42 LMMSE denoiser for any innovation law with the same first two moments — so single-
 43 Gaussian “message error” and “model error” are the same quantity.
- 44 • **Learning the kernel by EM through exact BP** (§3), with three structural properties that
 45 make it cheap: an exact E-step, a sufficient statistic that compresses the whole E-step into
 46 one matrix, and a gradient from Fisher’s identity that requires no automatic differentiation
 47 through the BP recursion.
- 48 • **A sample-efficiency comparison against trained denoisers** (§4), with the capacity, yard-
 49 stick and baseline-competence controls that make it a measurement rather than a favourable
 50 setup — and with the two facts that cut against it stated in the same place.

51 2 The exact denoiser

52 Let $a \in \mathbb{R}^n$ be drawn from $p_0(a) = \mu(a_1) \prod_{i=2}^n K(a_i | a_{i-1})$ with $K(a' | a) = \varphi(a' - \rho a)$ for an
 53 innovation density φ of mean zero and variance q . Noising is coordinatewise Ornstein–Uhlenbeck,
 54 so

$$p_t(a | x) \propto \mu(a_1) \prod_{i=2}^n K(a_i | a_{i-1}) \prod_{i=1}^n \ell_t(x_i | a_i), \quad \ell_t(x_i | a_i) \propto \exp\left(-\frac{(x_i - \alpha_t a_i)^2}{2\Delta_t}\right). \quad (2)$$

55 The likelihood factors are unary. They change the node potentials and leave the edge set untouched,
 56 so the posterior factor graph is the same chain as the prior’s, and the two sum-product sweeps

$$L_i(a_i) \propto \int K(a_i | a_{i-1}) \ell_t(x_{i-1} | a_{i-1}) L_{i-1}(a_{i-1}) da_{i-1}, \quad R_i(a_i) = \int K(a_{i+1} | a_i) \ell_t(x_{i+1} | a_{i+1}) R_{i+1}(a_{i+1}) da_{i+1} \quad (3)$$

57 give exact marginals, hence the exact posterior mean and, through (1), the exact score. We represent
 58 messages on a fixed quadrature grid; the discretisation is spectrally accurate and its error (9.2×10^{-15}
 59 at our working resolution) is four orders of magnitude below any learning error reported here.

60 **The Gaussian closure is the LMMSE denoiser, exactly.** A natural approximation propagates
 61 only the first two moments of each message. Because a Gaussian message times a Gaussian
 62 likelihood is Gaussian, and the transition step maps $\mathcal{N}(m, v)$ to a law with first two moments
 63 $(\rho m, \rho^2 v + q)$ regardless of the shape of φ , moment-matched BP discards every property of the
 64 innovation beyond its variance. It therefore returns precisely the linear minimum-mean-square esti-
 65 mator $\alpha_t \Sigma_0 \Sigma_t^{-1} x$ of the covariance-matched Gaussian model. We verify this to 6×10^{-16} across
 66 five innovation families. The consequence is that at the single-Gaussian level there is no distinction
 67 between message-representation error and model error — a distinction that becomes real only for
 68 richer message families, and which we separate elsewhere.

69 The size of that gap locates where non-Gaussianity matters. Against exact BP, the closure’s relative
 70 score error is largest at *small* t and decays monotonically: for a Laplace chain at $\rho = 0.85$ it runs
 71 from 0.20 at $t = 0.08$ to 9.2×10^{-5} at $t = 2.4$. Large t Gaussianises everything, so the prior’s shape
 72 survives only where the likelihood is weak.

73 3 Learning the kernel

74 We now suppose K unknown and observe only noised sequences. Writing θ for the kernel param-
 75 eters, EM alternates an E-step that computes posterior expectations under the current θ with an
 76 M-step that maximises the expected complete-data log-likelihood. Three properties of this instance
 77 matter.

The E-step is exact. In a generic latent-variable model the E-step is variational and the monotone-ascent guarantee is lost with it. Here the posterior is a chain for the reason given in §2, so BP returns exact single-site *and pairwise* marginals. Nothing is approximated but the message representation. Empirically the monotonicity violation is exactly zero in every run we report, which is the sharpest available test of the whole pipeline: an error anywhere in the recursion, the pairwise accumulation, or the M-step generically breaks it.

The whole E-step is one matrix. All the M-step needs is

$$\Xi_{kl} = \sum_{\text{sequences}} \sum_{i=2}^n \Pr(a_{i-1} = u_k, a_i = u_l \mid x), \quad (4)$$

the continuum analogue of Baum–Welch’s expected transition counts. It is sufficient and independent of how K_θ is parameterised, so the M-step never revisits the data, its cost is independent of the number of sequences, observations at different noise levels simply add into one Ξ , and changing kernel family changes only the M-step.

No automatic differentiation is required. Fisher’s identity gives

$$\nabla_\theta \mathcal{L}(\theta) = \langle \Xi(\theta), \nabla_\theta \log K_\theta \rangle, \quad (5)$$

the exact gradient of the *marginal* log-likelihood from a single BP pass. BP is not a computation graph to backpropagate through; it is the oracle that supplies the expectation. We verify this against finite differences of the exact evidence to $\sim 10^{-9}$. The implementation is plain numerical linear algebra throughout.

What noising costs. Identifiability is never destroyed by noising: the characteristic function satisfies $\varphi_{t,\theta}(\xi) = \varphi_\theta(\alpha_t \xi) \exp(-\Delta_t \|\xi\|^2/2)$, the Gaussian factor has no zeros, and $\alpha_t > 0$ for every finite t , so $p_{t,\theta} = p_{t,\theta'}$ forces $K_\theta = K_{\theta'}$. What degrades is *information*: the map is injective but ill-conditioned, this being Gaussian deconvolution. Measured per sequence, the Fisher information for the innovation scale falls by $142\times$ between $t = 0.05$ and $t = 1.6$, while that for the correlation falls by $26\times$. Non-Gaussianity is the first thing the channel destroys.

Estimator. We report a mixture-innovation kernel: $K(a' \mid a) = \sum_c \pi_c \mathcal{N}(a' - \rho a; \mu_c, \sigma_c^2)$, which retains the linear autoregression but leaves the innovation law essentially free. It has never seen the density it must recover. Fitted to noisy observations only, it returns the innovation variance essentially exactly and recovers the excess kurtosis of a Laplace innovation (3.0) monotonically in the component count, reaching 2.71 at eight components.

4 Results

All comparisons use a Laplace chain, $n = 32$, $\rho = 0.85$, with denoisers scored against exact BP under the true prior on held-out data. The reference itself is validated against importance sampling to 1.1×10^{-3} relative over six million samples, so the yardstick is not assumed.

4.1 Sample efficiency

Table 1 reports relative score error averaged over the noise schedule, over six independent training seeds, with the test set and its reference held fixed across seeds.

EM–BP on 32 sequences (0.0651 ± 0.0031) is better than the network on 4096 (0.1643 ± 0.0009) by a factor of 2.5 with non-overlapping intervals, so the network requires more than $128\times$ the data to match it — a lower bound, since 32 was the smallest budget tried and the network never caught up.

Three controls. *Capacity is not the explanation*: over 24 configurations from 3.2k to 297k parameters the best network reaches 0.208 against EM–BP’s 0.022, and error *degrades* for the largest models, so the gap is overfitting at this data budget rather than a badly chosen size. *The baseline can learn*: trained on a Gaussian chain where the exact denoiser is closed-form and linear, given four

Table 1: EM through exact BP (13 parameters) against a denoising network (25,248), at matched training-set size. Mean over six seeds, ± 1 standard error.

sequences	network	EM-BP	ratio
32	0.6363 ± 0.0043	0.0651 ± 0.0031	$9.8\times$
128	0.5117 ± 0.0025	0.0333 ± 0.0035	$15.4\times$
512	0.2813 ± 0.0013	0.0211 ± 0.0022	$13.3\times$
1024	0.2154 ± 0.0009	0.0182 ± 0.0021	$11.8\times$
2048	0.1756 ± 0.0009	0.0142 ± 0.0009	$12.4\times$
4096	0.1643 ± 0.0009	0.0124 ± 0.0005	$13.2\times$

Table 2: Against a locality-respecting architecture. Mean relative score error over five noise levels, four seeds, ± 1 standard error.

sequences	EM-BP (13)	global MLP (25,505)	local CNN (6,081)
128	0.0368 ± 0.0067	0.3595 ± 0.0044	0.0774 ± 0.0024
512	0.0174 ± 0.0017	0.1794 ± 0.0035	0.0570 ± 0.0007
2048	0.0123 ± 0.0009	0.1246 ± 0.0027	0.0505 ± 0.0015

times the data and twice the steps, it reaches 0.099 — the implementation is sound and the gap is a sample-efficiency effect. *Both parameterisations are trained:* predicting the noise wins at low t and predicting the clean signal at high t , by large factors, so we report the better of the two at every level and the result rests on neither.

4.2 What a structured architecture recovers

The comparison above is against a fully connected network carrying no locality prior, which is the natural first baseline but not the strongest one. We repeat it against a weight-shared window predictor — a one-dimensional convolution whose receptive-field radius and parameterisation are both chosen by oracle — over four seeds.

Roughly 60% of the deficit of the fully connected network was its architecture rather than the estimator: the margin falls from $\sim 10\times$ to 2.10 ± 0.39 at 128 sequences. But the remaining margin *grows* with data — $2.10 \rightarrow 3.27 \rightarrow 4.11 \pm 0.32$ — and the mechanism is visible in the columns. EM-BP falls by a factor of 3.0 over a $16\times$ increase in data while the CNN falls by 1.5 and is flattening. The structured network converges towards a floor set by its approximation error; the estimator keeps paying down parametric error. **The natural expectation that more data closes the gap is wrong here**, and the honest headline against a well-chosen architecture is $4.1\times$ and rising.

4.3 The advantage is tied to a small parametric dimension

Replacing the continuous chain by a discrete alphabet removes every discretisation caveat at once — BP becomes exact up to round-off, verified against explicit enumeration to 4.4×10^{-15} — and reproduces the same $\geq 64\times$ data-efficiency gap with zero grid error. It also shows the limit of the effect. Sweeping the alphabet size, the advantage over the best network at 512 sequences falls from $15.4\times$ at six parameters to $4.4\times$ at 132. Nothing here suggests a structured estimator retains its advantage once its parameter count approaches the network’s.

5 Limitations

Inference cost is the honest loss. Grid BP is $O(nM^2)$ per sequence per evaluation against a few matrix products for a network; we measure a $211\text{--}320\times$ slowdown. Because reverse diffusion calls the denoiser at every integration step, this is the weakest point of the result, and distillation of the fitted denoiser — not attempted here — is the obvious remedy. **Training is slower too:** EM’s cost grows linearly in the number of sequences while the network’s is fixed by its step count, so the advantage is in data, not wall-clock. **Everything rests on the chain assumption:** under a non-Markov prior the estimator is misspecified in a way no amount of data repairs. **One family,**

151 **one length, one correlation. And the comparisons above are pointwise**, whereas a diffusion
152 model is judged on what it generates; score error and sample quality are related by an integration
153 that accumulates error unevenly across the schedule, and we regard the sampling comparison as
154 necessary to complete the claim rather than as a formality.

155 **6 Conclusion**

156 A data model whose exact score is computable, and non-trivially so, makes it possible to ask what a
157 denoising network’s error is made of. On this model an estimator with thirteen parameters, learned
158 without ever seeing a clean sample and without differentiating through the inference algorithm,
159 reaches an accuracy that a conventional denoiser does not reach with two orders of magnitude more
160 data — and retains a growing advantage even against an architecture built with the right inductive
161 bias. The cost is inference time, and the scope is a single structured family. We take the useful
162 reading to be less that structure beats scale, and more that when the structure of the data is known,
163 the quantity that has to be estimated can be very small indeed.